

ENGR 102 Summer Bridge

Day 14 Student Quiz

List Indexing, Mutation, and Final-State Summaries

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace mutation and index-dependent updates.

4 points

```
values = [4, -2, 7, -1]
changed = 0

for index in range(len(values)):
    if values[index] < 0:
        values[index] = 0
        changed = changed + 1
    values[index] = values[index] + index

print(values, changed)
```

Give the list and changed state after every index, then write exact output.

Question 2. Repair a loop that changes only its temporary value.

4 points

```
values = [-2, 4, -1]
total = 0

for value in values:
    if value < 0:
        value = 0
    total = total + value

print(values, total)
```

Give actual output, explain why the list does not change, and rewrite the loop header/body to mutate positions.

Question 3. Program construction

12 points

Write one complete Python program that satisfies every requirement.

- Read threshold as an integer and use readings = [12, -3, 7, 20, 7].
- Loop by index and replace every negative value with 0.
- Using final values, compute total and the count at or above threshold.
- Find the index of the first largest final value without max or index.
- Print Final, Total, At or above, and Largest index on separate labeled lines.

[illegible]

Test evidence / expected result